

STM32™ ultra-low-power at 100 DMIPS with DSP and FPU



Meet our new budget friendly ultra-low-power STM32L4 MCUs

84 uA/MHz in RUN mode down to 8 nA in Shutdown



Our IoT Roadshow Dates

Bangalore	6th December 2016
Mumbai	8th December 2016
Delhi	15 December 2016

ULTRA-LOW-POWER EXCELLENCE WITH PERFORMANCE

- The STM32L4 microcontroller is based on a new ultra-low-power platform featuring Flex Power Control which extends flexibility to reach optimized power consumptions: with an EEMBC ULPBench score of 177 ULPBench™-CP, the STM32L4 Outperforms the market in the ultra-low-power domain.
- Offering up to 1 Mbyte of Flash (dual bank) memory and 128 Kbytes of SRAM, the STM32L4 unleashes the ARM Cortex-M4 power efficiency with floating point unit (FPU) and DSP instructions. It delivers 100 DMIPS / 273 CoreMark thanks to the ST ART Accelerator™ at 80 MHz. The entire system performance is optimized using a multi-AHB bus matrix and DMA controllers.

ST has all the ingredients for the IoT



- Sensors & Actuators
- Microcontrollers & Memories
- Ultra-low power connectivity
- Analog & Mixed Signal components
- Smart energy management

OUTSTANDING LOW-POWER MODES

Wake-up time	VBAT	2 nA / 200 nA*
250 µs	Shutdown	8 nA / 200 nA*
14 µs	Standby	28 nA / 280 nA*
14 µs	Standby + 16-Kbytes RAM	200 nA / 450 nA*
5 µs	Stop 2 (full retention)	1.0 µA / 1.28 µA*
4 µs	Stop 1 (full retention)	4.3 µA / 4.7 µA*
6 cycles	Sleep	26 µA / MHz
	Run at 24 MHz	84 µA / MHz
	Run at 80 MHz	102 µA / MHz

* without RTC / with RTC

Key Applications: • IOT • Smart Metering • Automation

Development as easy as 1 2 3



A full set of evaluation boards enables flexible prototyping as well as full STM32L4 evaluation.

Commercial part numbers:

- NUCLEO-L432KC (32 pins)
- NUCLEO-476RG (64 pins)
- STM32L476G-DISCO
- STM32L476G-EVAL



SOFTWARE

STM32CubeMX enables fast development thanks to its MCU clock configurator, power consumption calculator and code generation tools.

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